Disentangling Emotional and Linguistic Cues in AI-Generated Voice

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Background

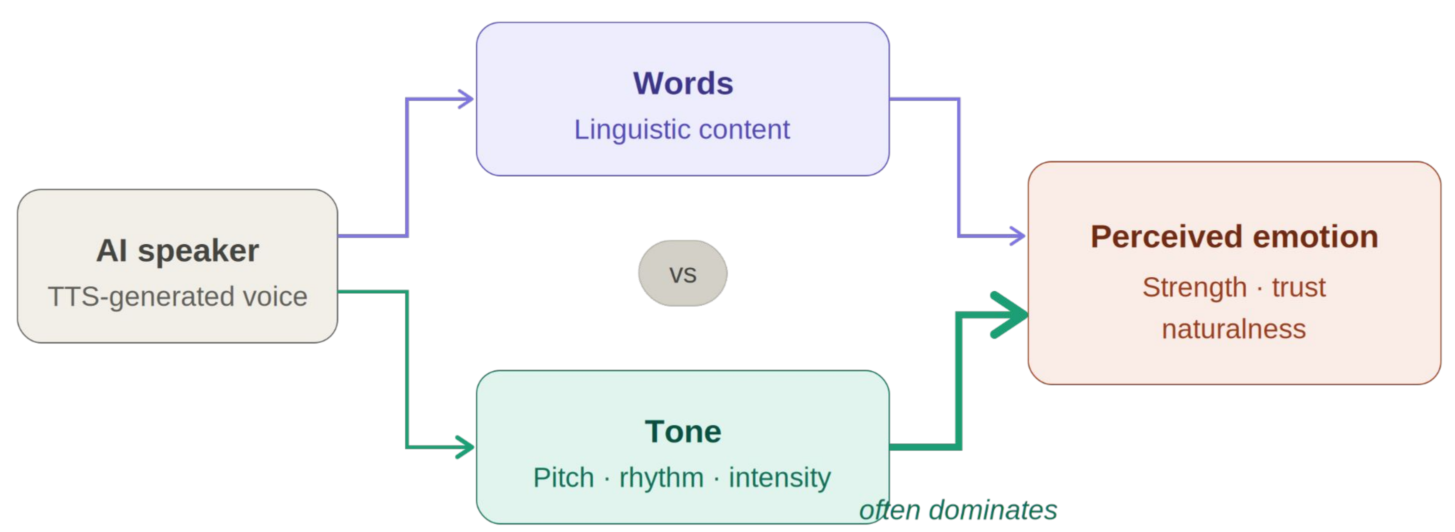
The Problem: “Synthetic Emotion”

- Modern AI voices (e.g. ElevenLabs, Amazon Polly) simulate emotion without feeling it. Users may misjudge intent based on tone alone.

Core Question: Do people rely more on **words** or **tone** when judging AI speech?

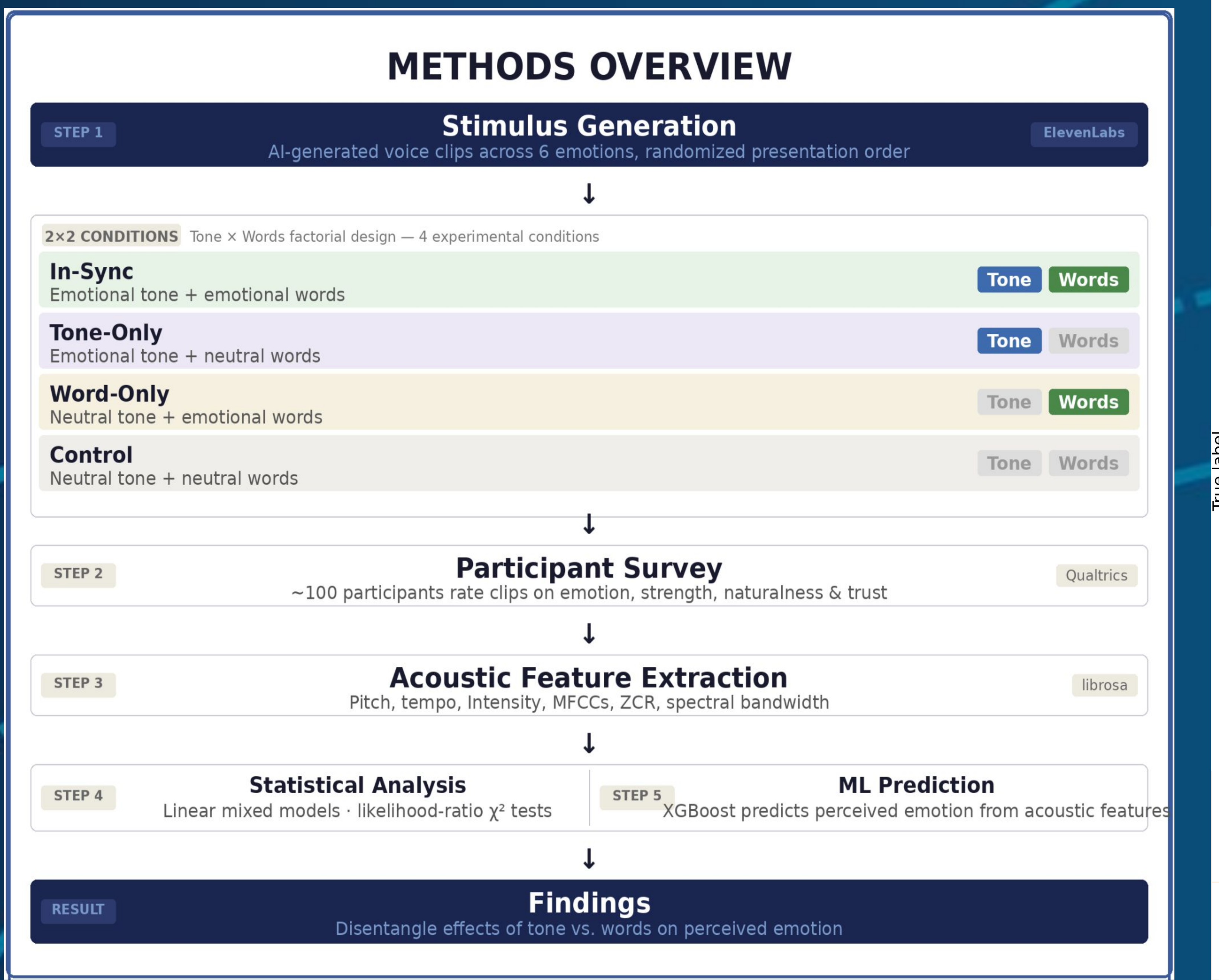
Why This Matters

- Perceived emotion shapes trust. Mismatches can mislead users. AI voice design should be transparent and user-centered.



Research Objectives

- Isolate whether tone or words more strongly drives perceived emotion
- Measure emotion strength, naturalness, and trust across 4 conditions
- Predict perceived emotion from acoustic features using machine learning



Dataset

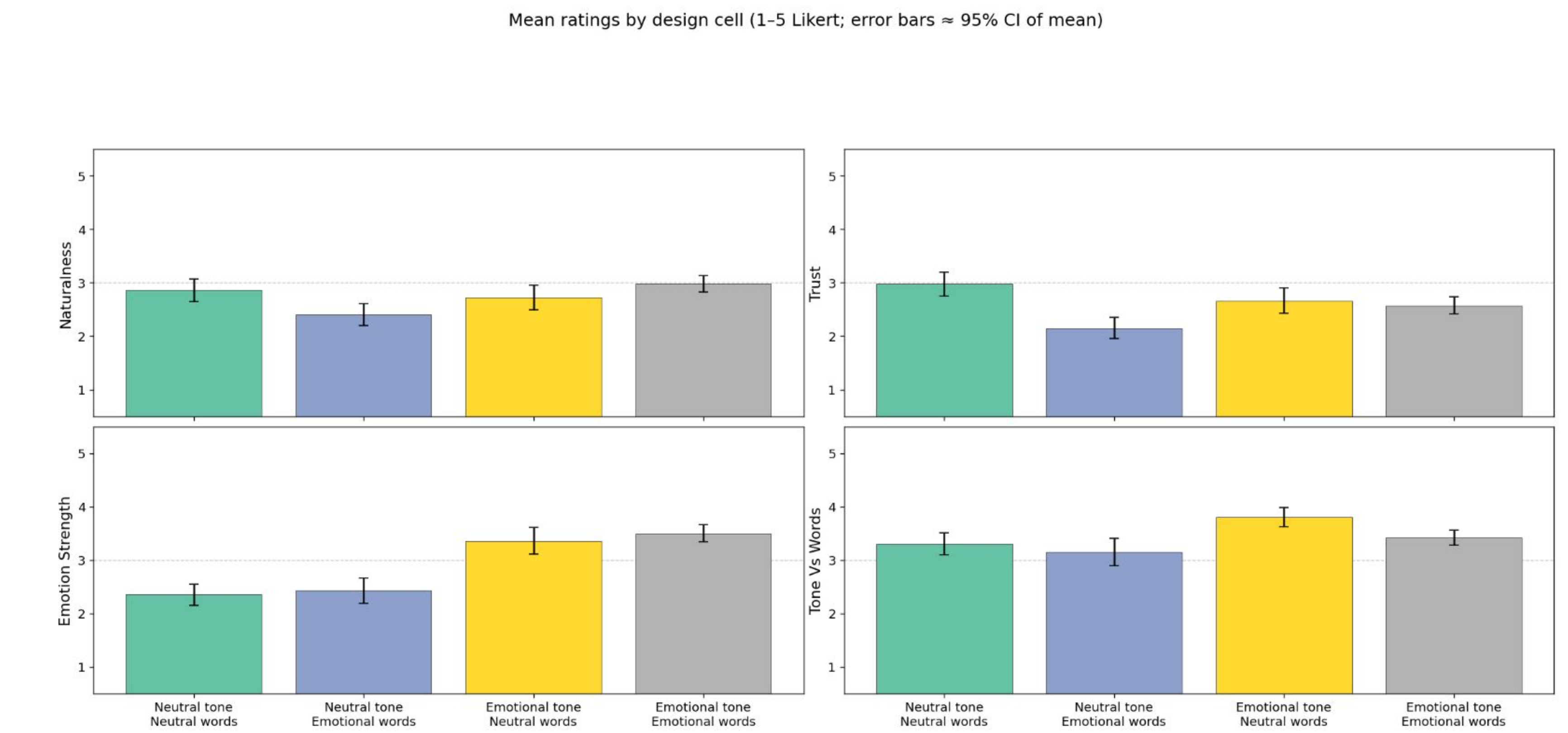
Data Sources

- 32 custom AI-generated speech clips across the 4 experimental conditions
- 102 participants, each rating 5 randomly assigned clips, 489 total ratings across all tone-word conditions (some incomplete responses)
- Acoustic features extracted via librosa: RMS, F0, ZCR, spectral centroid/rolloff/bandwidth, and MFCCs 1–13
- Survey responses mapped to clip IDs via randomized tracking

Results

Descriptive Pattern (Means by Condition)

- Emotion strength and naturalness **highest** when tone and words were both emotional
- Trust **highest** when tone and words were both neutral
- Tone-vs-words ratings above midpoint in all conditions; participants **leaned on tone over words**

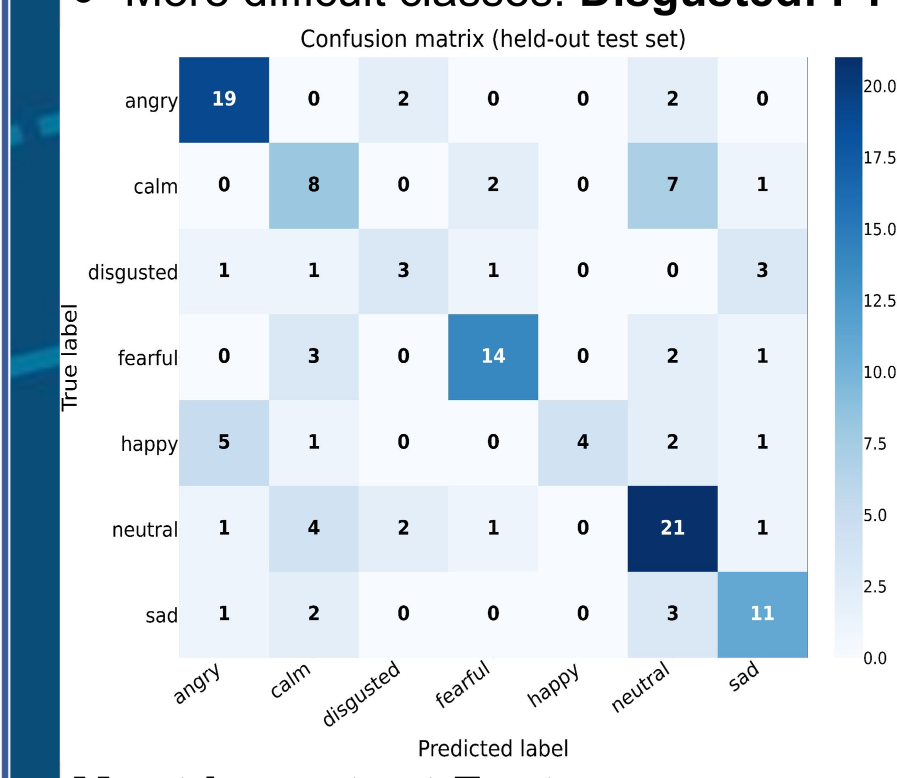


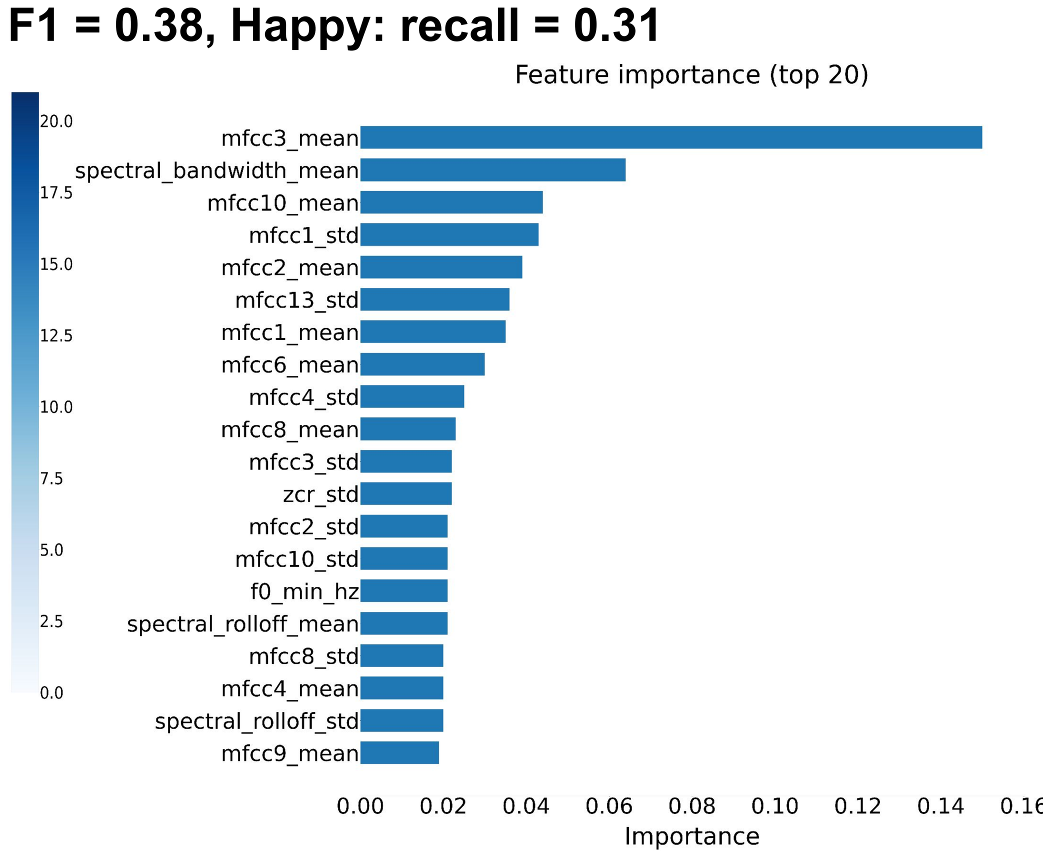
Statistical Evidence Across Conditions

- Naturalness differs by condition: $\chi^2(3) = 13.20$, $p = 0.004$
- Trust differs by condition: $\chi^2(3) = 21.16$, $p < 0.001$
- Emotion strength differs strongly by condition: $\chi^2(3) = 88.87$, $p < 0.001$
- Tone-vs-words weighting differs by condition: $\chi^2(3) = 8.98$, $p = 0.030$

Machine Learning Results (Perceived Emotion Prediction)

- Model: **XGBoost** classifier (grouped split by participant ID)
- Test accuracy: **61.5%**
- Target labels: 7 emotions (angry, calm, disgusted, fearful, happy, neutral, sad)
- Best-performing classes: **Angry: F1 = 0.76, Fearful: F1 = 0.74**
- More difficult classes: **Disgusted: F1 = 0.38, Happy: recall = 0.31**



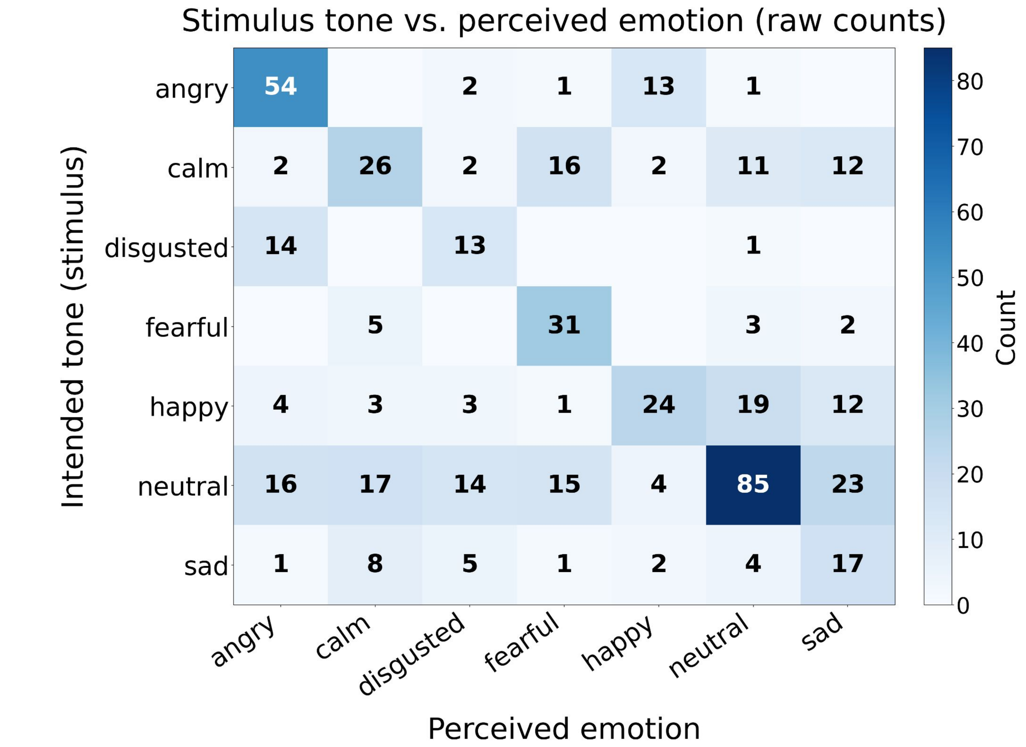


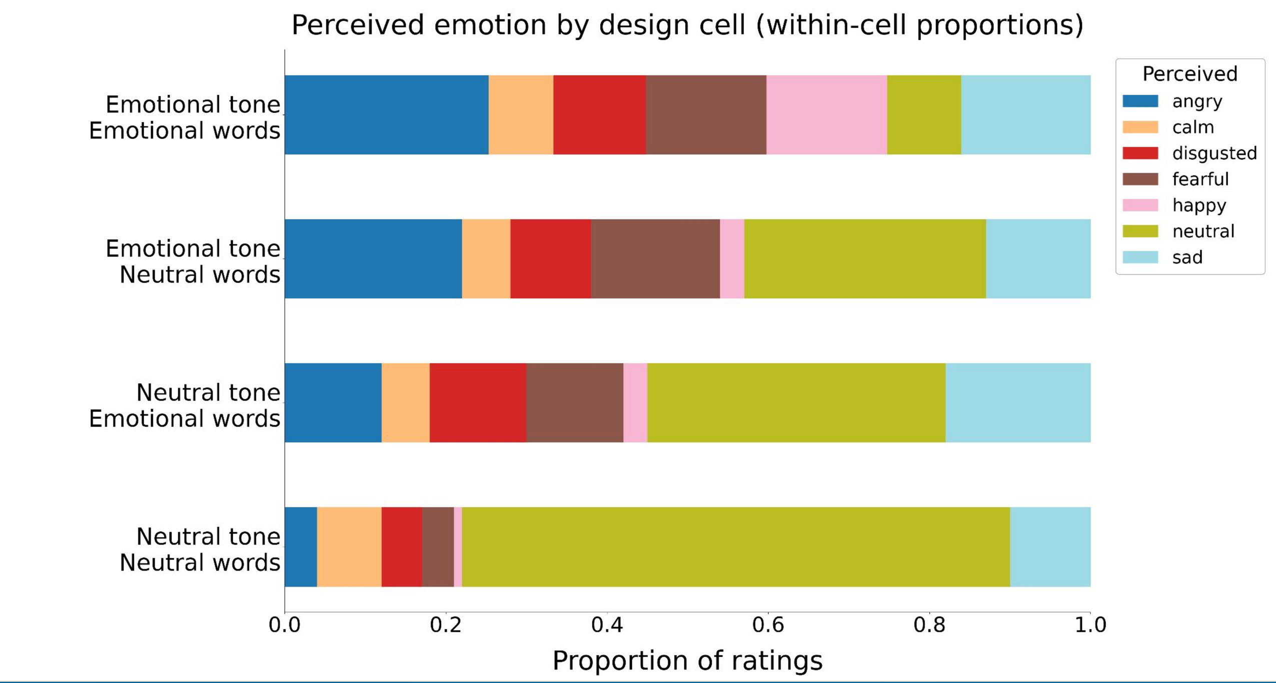
Most Important Features:

- **MFCC**: Spectral shape of speech capturing timbre and voice texture
- **ZCR:** Captures vocal roughness or sharpness, which can reflect tense or harsh delivery
- **Spectral Bandwidth**: Measures how spread out frequencies are in the voice signal
- **F0 (Fundamental Frequency)**: Captures pitch height and pitch variability

Conclusion

- Prosody and linguistic content both shape emotion perception, but their influence depends on the outcome being measured
- Emotion strength was **most sensitive to condition** showing that cue manipulation strongly affects how intensely emotion is perceived
- Aligned cues (emotional tone + emotional words) produced the strongest emotional responses and highest naturalness, suggesting cue alignment increases both affective impact and perceived realism
- **Neutral cues produced the highest trust,** implying users may prefer clarity over emotional expressiveness when judging reliability
- XGBoost reached **61.5% accuracy across 7 classes** from acoustic features, bounded by identical features mapping to different emotion labels.
- Angry and Fearful were easiest to classify; Disgusted and Happy most confusable, indicating uneven acoustic signatures across emotions.
- The heatmap shows intended and perceived emotions often differ, highlighting predictable misclassification in AI-generated speech
- Overall, findings support our main claim: AI emotional speech perception can be **disentangled**, and design choices should **balance emotional expressiveness, naturalness**, and trust based on use case





Limitations and Acknowledgements

Limitations

- Small sample: ~100 participants may limit power for statistical tests
- Limited diversity: Results may not generalize broadly to other demographics
- Forced labels: Fixed categories miss and oversimplify nuanced emotions
- Cue overlap: Tone and words are hard to fully isolate and may covary
- Low realism: Short clips lack real conversation context and interactions
- Non-causal ML: Feature importance shows association, not causation

Acknowledgements

- Research mentor Zhishang: Guidance on design and analysis.
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